

- Question 1 Consider the equation $x^4 = x^3 + 10$. We want to use the bisection method to solve this equation

1. Reformulate solving this equation as a root-finding problem.

We can reformulate the problem as a root finding one by finding the root of the function $f(x) = x^4 - x^3 - 10$, and for the sake of optimization the function can be written as $f(x) = x^3(x - 1) - 10$

2. Find an interval $[a, b]$ of length one that contains a solution of this equation. Explain why your interval satisfies the condition.

Note that $f(3) = 44$ and $f(2) = -2$. Therefore by the intermediate value theorem we have that there exists $c \in [a, b]$ such that $f(c) = 0$

3. Starting with your interval $[a, b]$, how many iterations of the bisection method are required to approximate the solution within an accuracy of 10^{-10} (aka, the absolute error is in $[-10^{-10}, 10^{-10}]$)?

Since our interval is of length 1, then $E_1 \leq \frac{1}{2}(3 - 2) = \frac{1}{2}$. Since $E_{n+1} \leq 2^{-1}E_n$ and $E_n \leq 2^{-n+1}E_1 = 2^{-n}$, then we have that

Question2fds10

$$2^n \geq 10^{10}$$

$$\log_2(2^n) \geq \log_2(10^{10})$$

$$n \geq 33.2192$$

$$2^{-n} \leq 10^{-\text{beginitemizeQuestion1fsdf}}$$

4. Question 2 fds

$$10$$

$$2^n \geq 10^{10}$$

$$\log_2(2^n) \geq \log_2(10^{10})$$

$$n \geq 33.2192$$

Therefore the minimum number of iterations required is 34.

Question 2 Assume that evaluating $f(x)$ for a given $x \in \mathbb{R}$ requires processor time t , while all other arithmetic operations take a negligible amount of time compared to t . Prove that performing n iterations of the secant method on $f(x)$ requires a total time of tn (aka, how much time do you need to calculate xn). (Hint: Previously computed function values can be reused in subsequent iterations.) Let x_0, x_1 be the starting guesses for the secant method with subsequent terms defined as:

$$x_{n+1} = x_n - f(x_n) \frac{x_n - x_{n-1}}{f(x_n) - f(x_{n-1})}.$$

To prove the claim that computing x_n takes tn time, we will be using induction. For the base case, note that to compute x_2 we have the formulabeginitemize

Question 1 fsdf

Question 2 fds

:

$$x_2 = x_1 - f(x_1) \frac{x_1 - x_0}{f(x_1) - f(x_0)}$$

Note that there are two unique calls to f , $f(x_0)$, $f(x_1)$, which have a combined run time of $2t$. Therefore we have demonstrated the base case. The induction hypothesis is for all $k \in \mathbb{N}$ if $k < n$ then the run time to compute x_k is tk . Now to compute x_n where $n > 2$, we have the formula:

$$x_n = x_{n-1} - f(x_{n-1}) \frac{x_{n-1} - x_{n-2}}{f(x_{n-1}) - f(x_{n-2})}.$$

Note that to compute x_{n-1} takes time $t(n-1)$ by the induction hypothesis. Additionally, to compute x_{n-1} the term $f(x_{n-2})$ was computed and saved, but not $f(x_{n-1})$, therefore we only must compute $f(x_{n-1})$. Since that will take exactly time t then we have a total run time for $t + t(n-1) = tn$.

Question 3 *On most computers, the computation of $\sqrt{a}$ is based on Newton's method for finding a root of $f(x) = x^2 - a$. Here, we assume $a > 0$. We want to use Newton's method to solve this root-finding problem.*

1. *What is the iteration formula of Newton's method for the above $f(x)$? The formula is*

$$\begin{aligned} x_n &= x_{n-1} - \frac{f(x_{n-1})}{f'(x_{n-1})} \\ x_n &= x_{n-1} - \frac{x_{n-1}^2 - a}{2x_{n-1}} \\ x_n &= \frac{a}{2x_{n-1}} - \frac{x_{n-1}}{2} \end{aligned}$$

2. *Note that $x^* = \sqrt{a}$ is a root of $f(x)$. Derive the error and relative error formulas*

- For the error we have that:

$$\begin{aligned} \sqrt{a} - x_{n+1} &= \sqrt{a} + \frac{x_n}{2} - \frac{a}{2x_n} \\ &= \sqrt{a} - \sqrt{a} + \frac{x_n}{2} + \frac{2}{2x_n} \sqrt{a} x_n - \frac{a}{2x_n} \\ &= \frac{x_n^2 + 2x_n \sqrt{a} - a}{2x_n} \\ &= \frac{-(a - 2x_n \sqrt{a} - x_n^2)}{2x_n} \\ &= \frac{-(\sqrt{a} - x_n)^2}{2x_n} \end{aligned}$$

- For the relative error we have that:

$$\begin{aligned}
 Rel(x_{n+1}) &= \frac{\sqrt{a} - x_{n+1}}{\sqrt{a}} \\
 &= \frac{-\frac{(\sqrt{a}-x_n)^2}{2x_n}}{\sqrt{a}} \\
 &= -\frac{\sqrt{a}(\sqrt{a}-x_n)^2}{2ax_n} \\
 &= -\frac{\sqrt{a}}{2x_n} \frac{(\sqrt{a}-x_n)^2}{a} \\
 &= -\frac{\sqrt{a}}{2x_n} \left(\frac{\sqrt{a}-x_n}{\sqrt{a}} \right)^2 \\
 &= -\frac{\sqrt{a}}{2x_n} Rel(x_n)^2
 \end{aligned}$$

3. For x_0 near a , the last formula becomes

$$Rel(x_{n+1}) \approx -\frac{1}{2} Rel(x_n)^2$$

Assume $Rel(x_0) = 0.1$, use this formula to estimate the relative error in x_1, x_2, x_3 .

- $Rel(x_1) = -0.005$
- $Rel(x_2) = -0.0000125$
- $Rel(x_3) = -0.000000000078125$

Question 4 Let the function $g(x) = \frac{x^2+4}{5}$

1. Find the fixed point(s) of $g(x)$

Note that the fixed point y will satisfy $y = g(y)$. Therefore we have the equation

$$\begin{aligned}
 y &= g(y) \\
 y &= \frac{y^2+4}{5} \\
 5y &= y^2+4 \\
 0 &= y^2-5y+4 \\
 0 &= (y-4)(y-1) \\
 y &= 1, 4
 \end{aligned}$$

Thus the fixed points are 1, 4.

2. Would the fixed point iteration, $x_{n+1} = g(x_n)$, converge to a fixed point in the interval $[0, 2]$ for all initial guesses $x_0 \in [0, 2]$? Please explain why.

The function $g(x)$ would converge in the interval $[0, 2]$ to 1 since $g'(x) = \frac{2}{5}x$, therefore for any possible guess x_0 within $[0, 2]$ we have that $0 \leq g'(x_0) \leq \frac{4}{5} < 1$, thus making g a contraction in that interval. Furthermore, they would all converge to 1 since for a delta $\delta = 1$ we have $[1 - \delta, 1 + \delta] = [0, 2]$. Thus by the theorem in the slides on february 14th 1 is the single fixed point which all values in the interval $[0, 2]$ converge to.

Question 5 Let $f(x) = \sin(x)$. Let $p_1(x), p_3(x), p_5(x)$ be the Taylor polynomials of degree 1, 3, and 5 for $f(x)$ and the point of approximation be 0 Note that the chain of derivatives for $\sin(0)$ goes 1, 0, -1, 0 reflecting the derivatives:

$$\sin \rightarrow \cos \rightarrow -\sin \rightarrow -\cos \rightarrow \sin$$

when evaluated at 0. Therefore the only terms that exists are the odd ones, thus giving the power series representation:

$$\sin(x) = \sum_{i=0}^{\infty} (-1)^i \frac{x^{2i+1}}{(2i+1)!}$$

where i represents the $i + 1$ th odd natural number. These facts will inform my solutions to the following problems:

1. What are $p_1(x), p_3(x), p_5(x)$?

- $p_1(x) = x$
- $p_3(x) = x - \frac{x^3}{6}$
- $p_5(x) = x - \frac{x^3}{6} + \frac{x^5}{120}$

2. What are $p_{2n-1}(x)$ and $p_{2n}(x)$ for interger $n \geq 1$?

- $p_{2n-1}(x) = \sum_{i=0}^{n-1} (-1)^i \frac{x^{2i+1}}{(2i+1)!}$
- $p_{2n}(x) = \sum_{i=0}^{n-1} (-1)^i \frac{x^{2i+1}}{(2i+1)!}$, note that there is no change since that $\frac{d}{dx^{2n}} \sin(0) = \pm \sin(0) = 0$

3. Compare the values of $f(x), p_1(x), p_3(x), p_5(x)$ for $x = \pm \frac{\pi}{2}, \pm \frac{\pi}{3}, \pm \frac{\pi}{4}$.

x	$p_1(x)$	$p_3(x)$	$p_5(x)$	$f(x)$
$\pm \frac{\pi}{2}$	± 1.5707963267948966	± 0.9248322292886504	± 1.0045248555348174	± 1.0
$\pm \frac{\pi}{3}$	± 1.0471975511965976	± 0.8558007815651173	± 0.8662952837868347	± 0.8660254037844386
$\pm \frac{\pi}{4}$	± 0.7853981633974483	± 0.7046526512091675	± 0.7071430457793603	± 0.7071067811865475